SPOTR™

RECON • DETECT • ALERT

Tactical Precision Camera Stream Viewer, Scope Reticle Customizer,
Micro-Change Thermal Heatmap, In-Memory Video Buffer & WebHID Controller Virtual Mouse.

BETA 1 RELEASE SPECIFICATION

OFFLINE PWA READY

1080p @ 60 FPS	WebHID BT Mouse	RAM Ring Buffer	Voice Directional	100% Client-Side
LOW LATENCY UVC STREAM	DUALSHOCK 4 TOUCHPAD & L2/R2	5S–60S INSTANT REPLAY	REAL-TIME TARGET ANNOUNCEMENTS	ZERO CLOUD DEPENDENCIES

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HARDWARE UVC Optical Scopes & Camera Input

SPOTR interfaces directly with standard USB Video Class (UVC) digital scopes, optical adapters, and webcams via the native HTML5 MediaDevices API.

- **Direct Plug & Play:** Hot-plugging a scope triggers automatic stream renegotiation without page refreshes.
- **Performance Profiles:** 1080p60 (recommended for scopes), 720p60 (low battery), or custom native capture.
- **Synthetic Test Signal:** If no camera is connected, SPOTR launches a synthetic test pattern card with calibrated hash lines to verify compositor and reticle settings.

LENS & ASPECT RATIO ENGINE

Canvas composition dynamically calculates letterbox/pillarbox geometry to preserve 1:1 true optical aspect ratios regardless of tablet display proportions.

INPUT BUS DualShock 4 Wireless Controller & Virtual Mouse

SPOTR features direct integration with Sony DualShock 4 controllers over Bluetooth utilizing the high-speed **WebHID API** (Report ID 0x11).

- **Touchpad Virtual Mouse:** Clicking the center touchpad toggles the high-contrast blue Virtual Mouse cursor. Gliding a finger smoothly moves the pointer across all toolbars and sliders.
- **Tactical Triggers (L2 / R2):**
  - `L2 Trigger`: Left Mouse Click & Drag (selects buttons, scrubs timelines, drags resizable PiP loupe).
  - `R2 Trigger`: Right Mouse Context Click.
- **Precision Joysticks:** Left Stick controls primary scope reticle aim with a calibrated 50% precision rate curve. Right Stick steers the digital PiP zoom window.

CONTROL MATRIX DualShock 4 & Bluetooth Ring Command Reference

CONTROL INPUT	PRIMARY VIEWPORT ACTION	INSTANT REPLAY / GALLERY MODE	RETICLE CUSTOMIZER MODE
`Touchpad Click`	Toggle Virtual Mouse Cursor ON / OFF	Toggle Virtual Mouse Cursor ON / OFF	Toggle Virtual Mouse Cursor ON / OFF
`Touchpad Glide`	Precision Mouse Pointer Translation	Precision Mouse Pointer Translation	Precision Mouse Pointer Translation
`L2 Trigger`	Left Mouse Click / Drag Floating Windows	Left Mouse Click / Drag Timeline Scrubber	Left Mouse Click / Drag Slider Knobs
`R2 Trigger`	Right Mouse Click	Right Mouse Click	Right Mouse Click
`Left Joystick`	Primary Reticle Aim (Precision 50% Speed)	—	Navigate Menu Items (Up/Down) & Adjust Values (Left/Right)

CONTROL INPUT	PRIMARY VIEWPORT ACTION	INSTANT REPLAY / GALLERY MODE	RETICLE CUSTOMIZER MODE
`Right Joystick`	Viewport Pan (when Zoom > 1.0x)	—	Sculpt Concentric Rings & Center Gap Diameter
`L1 / R1 Bumpers`	Main Viewport Digital Zoom (− / +)	Playback Video Zoom (− / +)	Quick Step Click Indents (−1 / +1)
`D-Pad ▲ / ▼`	Reticle FFP Subtension Spacing (Wider +5% / Tighter −5%)	—	Cycle Selected Setting / Drawer Focus
`D-Pad ◄ / ►`	Cyclic Quick-Tool Selector (Step Prev / Next Tool)	Step Backward / Forward (1 Frame Forensic Review)	Decrease / Increase Active Value Slider
`Cross (✕)`	Toggle Active Cyclic Quick Tool (ON / OFF)	Play / Pause Video Playback	Toggle Numbered Subtension Hashmark Labels
`Circle (◯)`	Declutter Screen / Universal Back to Live Feed	Close Replay & Return to Live Stream	Exit Customizer Screen
`Square (□)`	Toggle Scope Reticle Overlay (ON / OFF)	Toggle 0.5x Slow-Motion Playback	Toggle Concentric Range Rings
`Triangle (△)`	Toggle Thermal Micro-Motion Heatmap Engine	Toggle Saved Clips Drawer Panel	Toggle Center Bounding Brackets
`Share Button`	Instant 20s Replay (Auto-Saves to Local Storage)	Export Video Clip	Reset Reticle to Preset Defaults
`Options Button`	Open / Close Settings & Controller Configurator	Exit to Live Stream	Exit to Live Stream
`L3 / R3 Stick Clicks`	L3 / R3: Cycle Night-Vision Color Schemes (◀ Prev / Next ▶)	—	L3: Audition Preset | R3: Reset Preset

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VIEW 1 Live Feed

60 FPS live video, micro-change thermal heatmap, floating draggable PiP loupe, customizable crosshairs, FFP subtension scaling, and cyclic quick-tool dock.

VIEW 2 Instant Replay

Immediate native-resolution playback of the in-memory RAM buffer. Scrub frame-by-frame or play at 0.25x / 0.5x slow-motion.

VIEW 3 Video Archive

Permanent local storage library in IndexedDB. Organize shot clips, review high-speed impacts, and export directly as standalone .webm.

VIEW 4 & 5 Reticle & Controller

Parametric reticle studio (mils/MOA click corrections) & full-screen Controller & Key Rebinding Matrix with live input glow diagnostics.

CONTROLS Floating Toolbars & Declutter Mode

- **Top-Left Control Strip:**
  - **Settings Gear:** Opens the central configuration modal (RAM meter, camera selection, controller mappings).
  - **Target POI Icon:** Toggles the resizable PiP Digital Zoom Loupe.
  - **Crosshair Reticle Icon:** Turns scope reticle overlay ON / OFF.
  - **Thermal Heatmap Flame:** Toggles 60 FPS micro-motion detection.
  - **Voice Announcement Speaker:** Quick-toggle for speech engine with active waves (cyan) vs muted slash (red).
- **Top-Left Zoom Pod:** Whole-viewport digital zoom controls (`-` , `1.0x` , `+` , `1:1`) with smooth touch and joystick panning.
- **1-Tap Declutter Mode:** Tapping anywhere on open screen space (or pressing `R3`) hides all toolbars and buttons, leaving an uninhibited, 100% full-screen sight picture.

OPTICS Digital Zoom Loupe (#pipCanvas)

The Picture-in-Picture (PiP) Loupe provides real-time digital magnification of any specific point of interest (POI) on the target without changing the primary field of view.

- **Independent Pan & Zoom:** Adjust loupe magnification from 2.0x to 16.0x independently of main viewport zoom.
- **Sub-Pixel Target Crosshair:** A high-contrast target marker indicates the exact optical coordinate currently magnified in the loupe.
- **Draggable & Resizable:** Drag the loupe anywhere on the screen by its titlebar. Drag the bottom-right or bottom-left corner handles to resize.
- **Hardware Steering:** Push the Right Joystick on your gamepad to smoothly track the loupe target across the target face.

APPEARANCE Tactical Night Themes & Global Interface Opacity

Customize UI chrome and reticle colors to match ambient lighting and target contrast: **Emerald Teal** (#2dd4bf), **Amber Gold** (#ffd166), **Ruby Red** (#f43f5e), **Night Blue** (#38bdf8), **Amethyst** (#a855f7), or full 2D HSL color selection.

Interface Opacity Slider (15% to 100%): Fades background chrome, borders, and buttons into near-invisibility (`--ui-opacity`), allowing spotting downrange without night-blindness or screen glare.

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DETECTION

Thermal Micro-Change Heatmap Engine

SPOTR features an advanced, low-latency micro-movement detection algorithm that highlights subtle changes (bullet impacts, target movement, wind drift) as a colored thermal glow overlay.

- **Downsampled Luma Analysis:** Downsamples the 1080p stream to a high-speed 160×90 luminance matrix at 60 FPS, calculating delta variance against an exponential moving average background.
- **Global Shift Rejection (>18%):** Automatically discards whole-frame camera movements (wind buffeting, scope recoil, tripod bumps) to prevent false alerts.
- **Adjustable Sensitivity (10% to 95%):** Fine-tune detection thresholds in Settings to spot minute paper tears at 1000 yards without triggering on heat mirage.
- **Decay Duration (0.5s to 10s):** Controls how long the visual impact heat bloom persists before fading.

SPEECH

Directional Voice Announcer (`speechEngine.js`)

When target movement or bullet impact is detected, SPOTR computes the geometric centroid vector relative to your scope crosshair and speaks the direction out loud using the browser's Web Speech API.

- **Directional Speech Cues:** Announces target quadrant (e.g. "*Target High Right*", "*Movement Low Left*") or exact clock coordinates (e.g. "*Impact 2 o'clock*").
- **Customizable Keyword Dictionary:** In Settings, operators can replace standard directional words with custom military or tactical keywords (e.g., replace "High" with "North" or "Elevation").
- **Intelligent Throttling:** An internal speech queue and rate-limiter prevents audio cascades and ensures only fresh detections are spoken.
- **Immediate Voice Mute:** Click the speaker icon in the top-left toolbar or press the Mute hotkey to instantly silence audio without disabling visual heatmap rendering.

VOCABULARY

Directional Speech Dictionary & Configuration Options

RELATIVE ANGLE	DEFAULT ANNOUNCEMENT	CUSTOM KEYWORD OPTION	OPERATIONAL USE CASE
337.5° – 22.5° (0°)	**"Target High"**	"North" / "Top" / "Elevation Up"	Impact high of point of aim; adjust turret down.
22.5° – 67.5° (45°)	**"Target High Right"**	"Two O'Clock" / "North-East"	Impact high right; windage left + elevation down.
67.5° – 112.5° (90°)	**"Target Right"**	"East" / "Three O'Clock" / "Windage Right"	Crosswind deflection; adjust windage turret left.
112.5° – 157.5° (135°)	**"Target Low Right"**	"Four O'Clock" / "South-East"	Impact low right; hold over and into wind.
157.5° – 202.5° (180°)	**"Target Low"**	"South" / "Bottom" / "Elevation Down"	Impact low; adjust elevation turret up.

RELATIVE ANGLE	DEFAULT ANNOUNCEMENT	CUSTOM KEYWORD OPTION	OPERATIONAL USE CASE
Reticle Center (<5% radius)	**"Direct Hit" / "Bullseye"**	"Center" / "Hit"	Point of impact matches zero point of aim.

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IN-MEMORY Native-Resolution Circular RAM Ring Buffer

Traditional video recording causes tablet thermal throttling and flash storage wear. SPOTR solves this by maintaining a **100% RAM-only rolling queue** (`memoryRingBuffer.js`) with zero disk writes during surveillance.

- **Configurable Duration:** Store from **5 to 60 seconds** of rolling historical video.
- **Configurable Buffer FPS:** Set buffer capture between **5 and 30 FPS** to optimize RAM footprint on low-memory devices (e.g., Chromebook Duet 3 with 4GB RAM).
- **Live RAM Budget Meter:** Real-time telemetric bar in Settings displays Current Used MB, Estimated Allocation, and Free Device RAM.
- **Automatic Memory Guardrails:** Proactively purges older frames if device memory pressure exceeds threshold, guaranteeing rock-solid stability without browser crashes.

PLAYBACK Instant Replay Viewport & Scrubbing

Capture critical shot strings instantly with a single button press or ring tap.

- **Instant Snapshot:** Press `Circle` on controller, double-tap the center ring button, or click **REPLAY** to freeze and inspect the circular buffer.
- **Automatic Silent Persistence:** Upon trigger, the snapshot is automatically cloned to permanent local IndexedDB storage in the background without stuttering playback.
- **Precision Timeline Scrubber:** Drag the timeline slider to locate the exact microsecond of bullet impact or target change.
- **Slow-Motion & Frame Step:** Play at **0.25x**, **0.5x**, or step frame-by-frame forward (`STEP +1`) and backward (`STEP -1`) for forensic ballistic review.

STORAGE Video Archive Library & High-Speed WebM Exporter

1. IndexedDB Local Storage

Clips are stored securely in browser sandboxed storage. Clips remain accessible across browser restarts and offline field sessions without cloud connectivity.

2. Fullscreen Analysis Player

Review saved clips in the full-screen Gallery Player. Features a collapsible side drawer with clip timestamps, frame counts, and duration badges.

3. One-Click WebM Export

The native export engine encodes recorded frames into a standard `.webm` video file, saving it directly to local disk for reporting, archiving, or sharing.

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RETICLE

Parametric Reticle Studio (Up to 30 Clicks) & Center Point Styles

Craft the ideal sighting overlay for any optics magnification or target geometry in View 4 (#screenReticle):

- **Extended 30-Click Elevation/Windage Ladder:** Display up to 30 clicks/indents per arm with major graduation labels (every 5 clicks) across vertical and horizontal axes for extreme long-range holdovers.
- **Optical Dead-Center Measurement:** All click distances and voice callout turret adjustments are calculated strictly from true optical dead-center (0, 0), completely independent of visual center gap diameter.
- **4 Selectable Center Aiming Symbols:**
  - **EMPTY:** Open unobstructed center viewing window.
  - **DOT:** Precision center aiming dot at the optical origin.
  - **CROSS:** Fine mini-crosshair intersection at dead-center.
  - **CHEVRON:** Tactical inverted caret (Λ) whose sharp apex points directly to the exact measurement pixel (0, 0).
- **Direct Controller Sculpting & Cycle:** Adjust click count, spacing, gap, stroke weight, or press `Key C` / Quick Tool dock to cycle center point styles live.

CALIBRATION

MOA & MRAD Turret Click Calculator

Eliminate sighting guesswork by calculating exact optical turret click offsets directly from observed bullet impacts:

- **Target Distance Input:** Enter engagement distance in Yards or Meters.
- **Turret Value Selection:** Select your scope turret click standard (1/4 MOA, 1/8 MOA, 1/2 MOA, 0.1 MRAD, or 0.05 MRAD).
- **Point of Impact (POI) vs Point of Aim (POA):** Measure target offset in millimeters or inches on the digital display.
- **Instant Turret Solution:** SPOTR calculates exact turret clicks (e.g. "*Elevation: +6 Clicks UP | Windage: -4 Clicks LEFT*") to achieve pinpoint zero.

DEPLOYMENT

Progressive Web App (PWA) Offline Installation Guide

SPOTR is built as a self-contained, offline-first Progressive Web App with zero external runtime dependencies. Precached by Service Worker `spotr-v36`:

ChromeOS / Chromebook

In Chrome, click the **Install** icon in the omnibox. Launches in standalone window mode with Window Controls Overlay.

Windows / PC

Open in Chrome/Edge, click **Settings > Apps > Install SPOTR**. Creates a desktop shortcut and starts like a native executable.

Android Tablets & Phones

Tap browser menu (⋮) > **Add to Home Screen**. Operates in full-screen immersive landscape mode.

Apple macOS & iPadOS

In Safari, tap **Share > Add to Home Screen**. Pre-caches all assets for completely offline field operation.